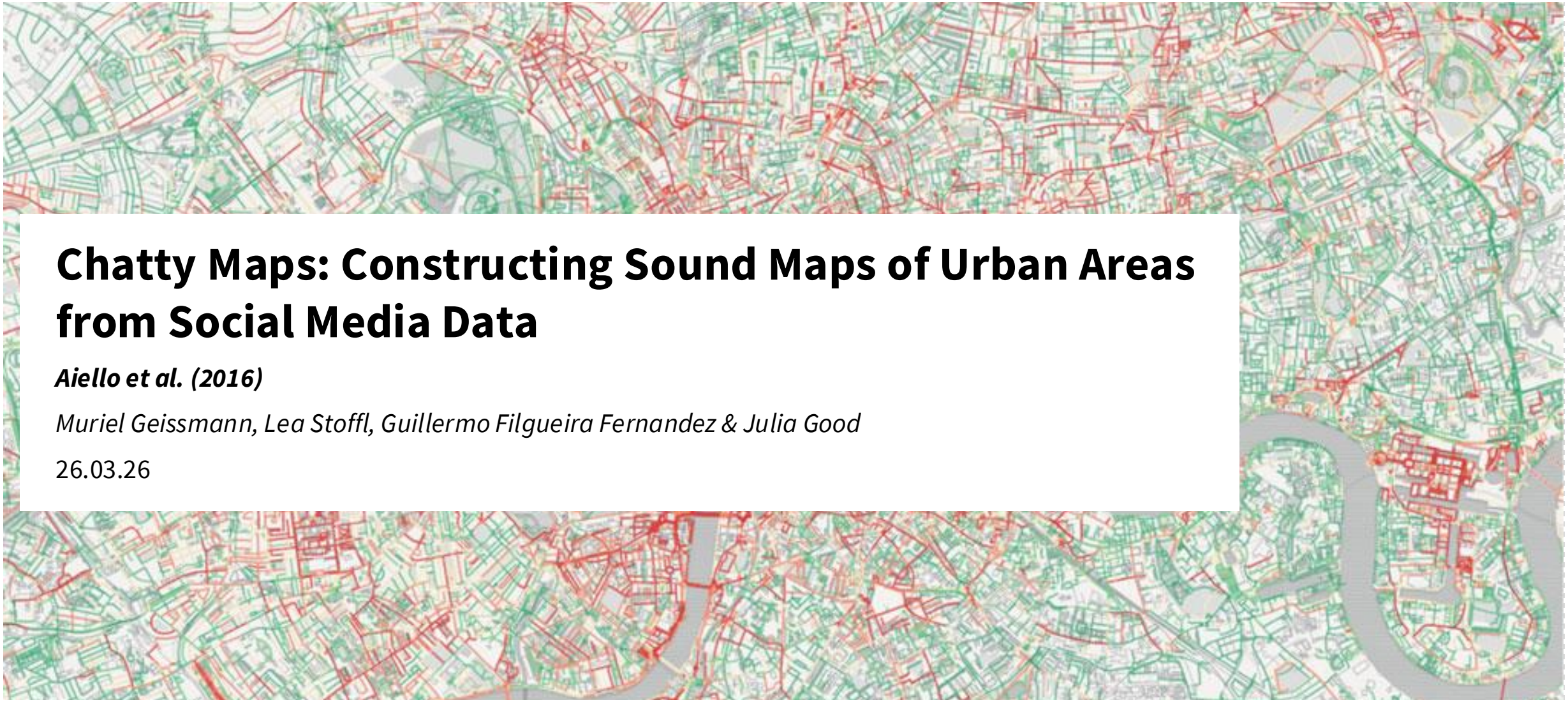


Chatty Maps: Constructing Sound Maps of Urban Areas from Social Media Data

Aiello et al. (2016)

Muriel Geissmann, Lea Stoffl, Guillermo Filgueira Fernandez & Julia Good

26.03.26

Research Questions

1. Can urban soundscapes be inferred from social media data, specifically from tags of geotagged photos? ?
2. Can these data be used to classify types of urban sounds (traffic, nature, people, music)?
3. How are different types of sounds associated with human emotions and perceptions of places?
4. Can social media data provide a useful alternative or complement to traditional noise pollution measurements?

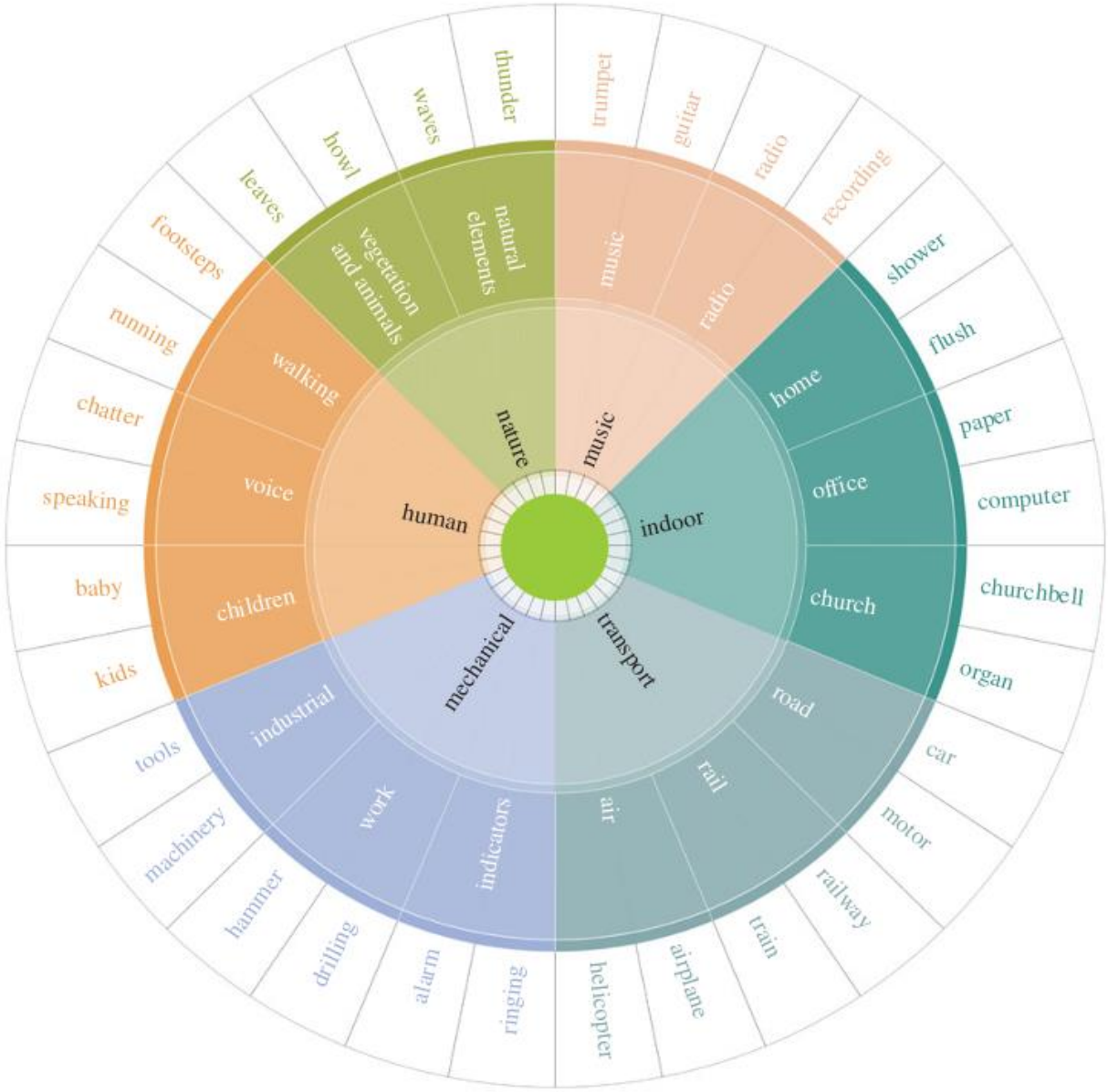
Data Sources

- Main dataset consists of geotagged Flickr photos
- For each photo we have: Photo ID, User ID, latitude and longitude and user-generated tags
- Sound-related dictionaries
- Entries from the word-emotion lexicon EmoLex
- Soundwalks conducted in the UK and Italy

- Additional data: noise pollution map for Barcelona

Data Structure

- Dictionary of sound-related words
- Cities divided into street segments
- calculate the frequency of sound related tags and set dominant sound category



Spatial Questions: Build the spatial sound layer

The first tasks convert scattered tags into a street-level analytical surface.

Geo-tagged Flickr photos
Input points

Match tags to sound words
Semantic filter

Aggregate to street segments
Main spatial unit

Group into 6 sound categories
Spatial classification

1

Where are different sounds located?

Map tags to OpenStreetMap street segments, then estimate each segment's sound profile.

2

Which sound types belong together?

Use spatial co-occurrence of tags to build categories such as transport, nature, music, and human.

Key idea: the street segment is the basic spatial unit, so the study becomes a city-wide comparison of places.

Spatial Questions: Test spatial patterns and validate them

The next tasks ask whether the mapped soundscape behaves like a real city.

3

Which sounds co-locate, and which are separated?

Correlate sound categories across street segments: for example, nature and transport should fall on different parts of the city.

4

How do sound profiles vary by street type?

Compare primary roads, residential streets, footways, pedestrian streets, and construction areas.

5

Do social-media sound maps match measured noise?

Validate segment sound profiles against Barcelona's official noise data.

These tasks show the map is not random: it reflects meaningful urban structure.

Nature **Transport**

Example spatial separation

Spatial Questions: From sound sources to human experience

The final tasks ask what those spatial sound patterns mean emotionally and perceptually.

6

How are emotions distributed across space?

Link each street segment's sound profile to emotional language in tags; music and human sounds align more with joy, traffic with fear and anger.

7

How are places likely to be perceived?

Use soundwalk survey results to estimate whether segments are calm, chaotic, vibrant, or monotonous.

Takeaway

From mapped sounds to mapped experience.

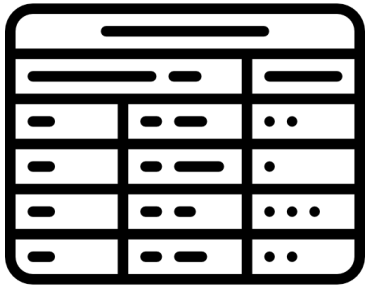
Chaotic **Vibrant**

Monotonous **Calm**

Replication of Study for Zurich: Research Questions

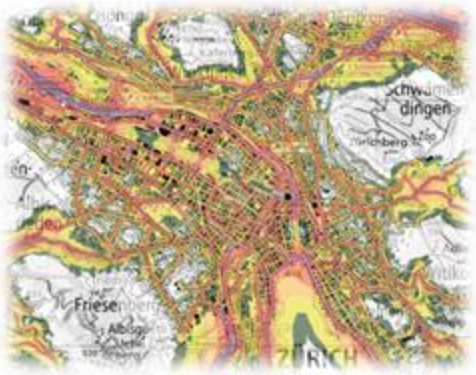
1. Can urban soundscapes in Zurich be inferred from geotagged Flickr tags?
2. According to the Flickr tags, which noise categories are dominant on different road sections in Zurich?
3. How strongly do Flickr-derived sound patterns correlate with official noise pollution datasets?
4. Can Flickr complement traditional urban sound monitoring?

Replication of Study for Zurich: Data Sources



Geotagged Photos:
Flickr posts

Streets:
Official Street Directory of Switzerland

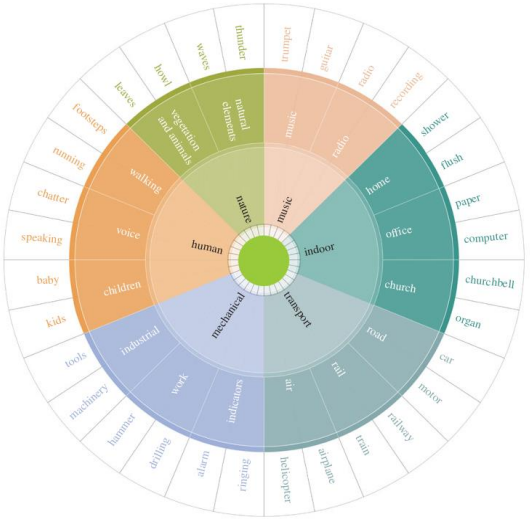


Noise Pollution Data:
Road Noise data of Zurich

Spatial boundary:
Municipal boundary of Zurich

Replication of Study for Zurich: Data Preprocessing and Workflow

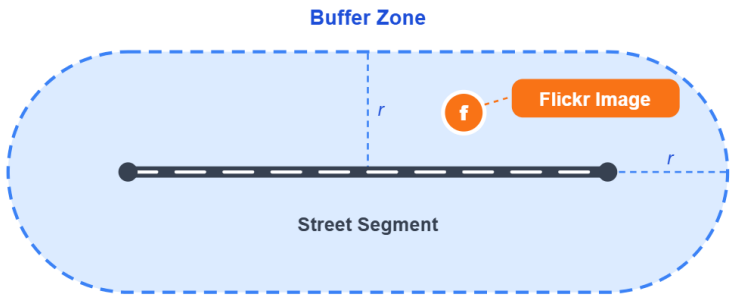
- Tag Preprocessing → filter out Flickr posts with no language-related tag
  → translate tags in the same language
- Take taxonomy of sound-related words from the paper
- Divide Zurich into street segments by street name and add a buffer

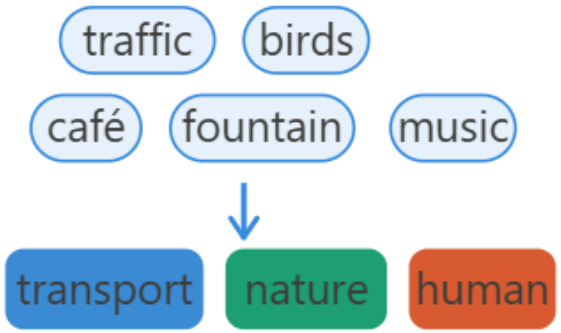


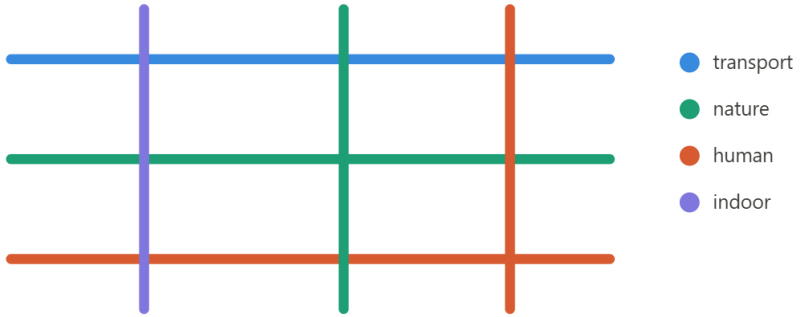
Category	Keywords
• Transport	• car, bus, train, tram, engine
• Nature	• bird, wind, water
• Human	• crowd, people, talking, kids
• Music	• music, concert, band
• Mechanical	• machine, construction
• Indoor	• Church, shower, computer

Replication of Study for Zurich: Workflow

Point in Polygon Spatial Join → Matching Tags to Sound Categories → Calculation of Dominant Sound Category → Visualization of Sound Categories







Images created by Claude

Replication of Study for Zurich: Workflow (Noise Pollution Validation)

- Compare sound category map with Zurich's traffic noise pollution data
- Correlation analysis: Do Flickr-derived categories align with measured traffic noise?

Expected:

- Transport-dominant streets → high noise
- Nature-dominant streets → low noise

Noise level	Transport	Human	Indoor	Mechanical	Nature
High (>65 dB)	+0.72	+0.31	+0.05	+0.28	−0.61
Medium (50–65 dB)	+0.24	+0.38	+0.42	+0.11	−0.08
Low (<50 dB)	−0.65	−0.12	+0.19	−0.22	+0.68

Correlation (r): strong positive (>0.5) weak positive (0.1–0.5) neutral (−0.1–0.1) weak negative (−0.5–−0.1) strong negative (<−0.5)

Table created by Claude